

Shot point notebook

Olympic Trap · ISSF Olympic 1 · 15 clays

What you have in your hands

A complete round, clay by clay, exactly as the shooter would see it from the station. For each clay the whole trajectory is drawn, with five moments marked on it, spread across the window in which the clay is still within reach, plus one that is already late.

At each moment it tells you where the clay is and how much lead you need to break it, measured as the eye sees it. It is meant for planning the round before you go to the ground: what you shoot first, how much you lead, and what happens to you if you are late.

How to read each sheet

- The large figure is the lead as the eye perceives it, perpendicular to the line of sight. Below it also appears in clay widths and in degrees.
- The real three-dimensional separation is in small print: it includes the depth, which the eye does not see, and that is why it is a much larger number.
- Every sheet shares the same scale, so leads are comparable from one clay to another. The header says how much the view has been turned and with what framing.
- The colour of each moment warns you about distance: within reach, at the limit, or already beyond useful range.

What it is not

- It is not a field measurement: these are figures from a teaching model, with its simplifications. The last page explains this in detail.
- It does not certify an installation or a trajectory. In competition the authorised officials always decide.
- It does not replace an instructor, the rules of the ground, or the legal requirements.

Generated with these settings

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral

Every lead in this notebook is valid only for that combination. With another cartridge, another rib, another choke or with wind, the figures change and this notebook stops being useful: you have to generate a new one.

You do that from the simulator, changing whatever you want and pressing «Generate notebook» again. On each sheet, the settings that are not the start-up ones appear highlighted in colour.

The project

Plato is a free simulator and book for learning clay target shooting, written by Roberto Canales Mora. The book explains the ground, safety, the disciplines and the geometry of lead; the simulator lets you repeat the same clay changing one single variable.

Project, contact and repository on GitHub:

www.robertocanales.com/proyectos#plato
rcanalescoder.github.io/Plato

What is on the following pages

- An index of the round: a table with station, machine, angle, height, fall and lead for each clay, and a plan view with all the trajectories together.
- Una ficha por plato, 15 en total, en el orden en que los verías durante la tirada.
- The notice of use and the licence, at the end. If you print only part of it, print that sheet too.

Notice: the figures in this notebook are an estimate from a teaching model, not a field measurement. Read the notice of use and the licence on the last page before taking it to the ground.

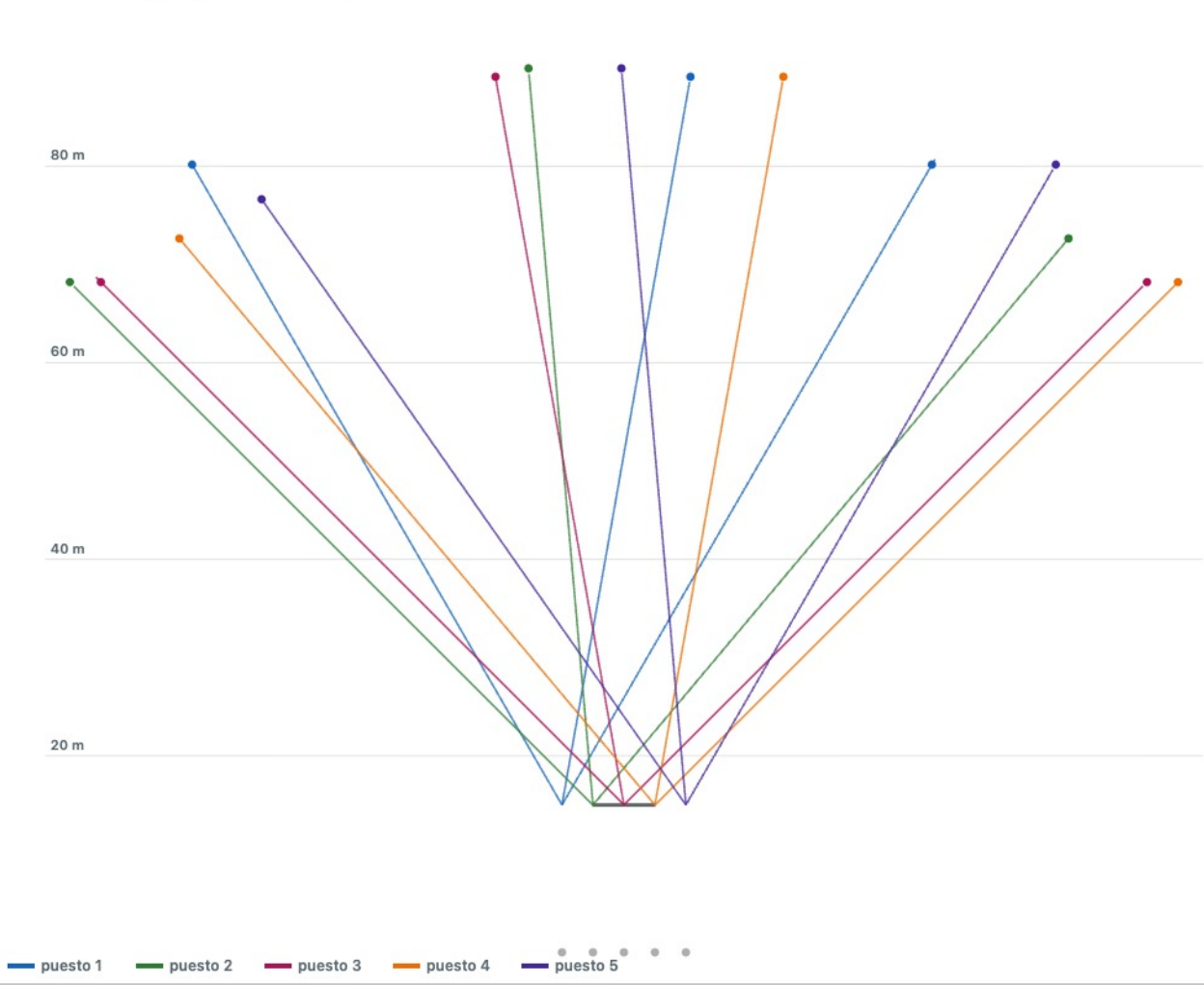
Shot point notebook

Olympic Trap · ISSF Olympic 1 · 15 clays

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral

#	station	mach.	angle	h. 10 m	fall	lead in the good window
1	1	1	+30°	2.5 m	76 m	89 cm · 8.1 platos · 1.01 s
2	1	2	+10°	1.5 m	76 m	45 cm · 4.1 platos · 0.76 s
3	1	3	-30°	3.0 m	76 m	79 cm · 7.2 platos · 1.10 s
4	2	4	+40°	1.7 m	76 m	145 cm · 13.2 platos · 0.86 s
5	2	5	-5°	2.0 m	76 m	20 cm · 1.8 platos · 0.87 s
6	2	6	-45°	2.0 m	76 m	143 cm · 13.0 platos · 0.95 s
7	3	7	+45°	2.0 m	76 m	143 cm · 13.0 platos · 0.95 s
8	3	8	-10°	3.0 m	76 m	28 cm · 2.6 platos · 1.06 s
9	3	9	-45°	2.8 m	76 m	116 cm · 10.6 platos · 1.11 s
10	4	10	+45°	2.3 m	76 m	131 cm · 11.9 platos · 1.02 s
11	4	11	+10°	1.8 m	76 m	39 cm · 3.6 platos · 0.83 s
12	4	12	-40°	1.6 m	76 m	152 cm · 13.8 platos · 0.84 s
13	5	13	+30°	2.0 m	76 m	102 cm · 9.3 platos · 0.91 s
14	5	14	-5°	1.5 m	76 m	25 cm · 2.2 platos · 0.76 s
15	5	15	-35°	3.0 m	76 m	91 cm · 8.3 platos · 1.11 s

Plan view · every clay in the scheme



The table uses the «good window» moment, the comfortable shot; each sheet details the other moments.

The lead is the one the shooter sees: the apparent separation between clay and aiming point, measured perpendicular to the line of sight.

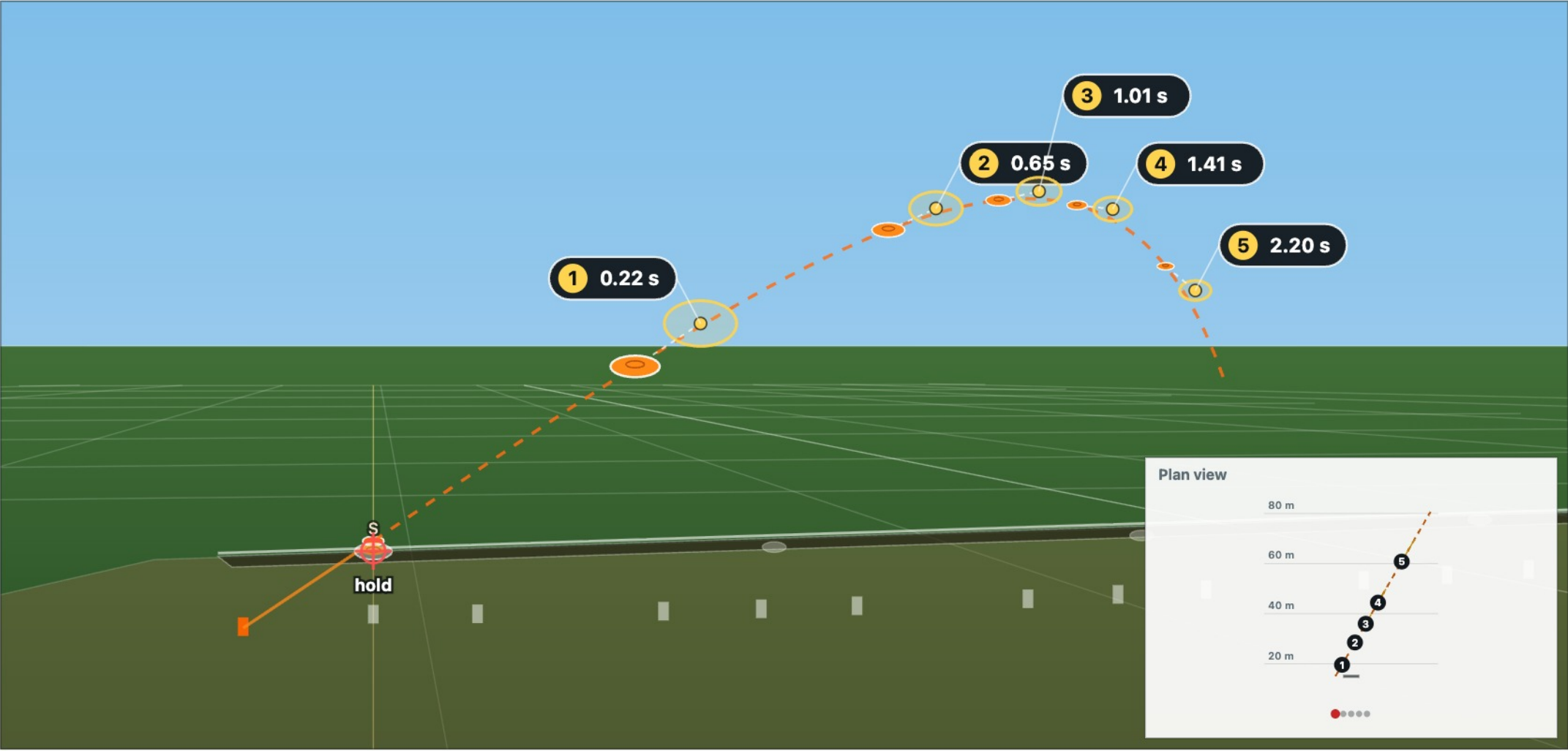
Red: the gun's hold point. Orange: the clay. Yellow: where to put the aiming point. Yellow ellipse: estimated pattern of the first shot.

Station 1 · Machine 1

angle +30° · height 2.5 m at 10 m · fall 76 m · 90 km/h · elevation 15.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned +12° horizontally and +5° vertically



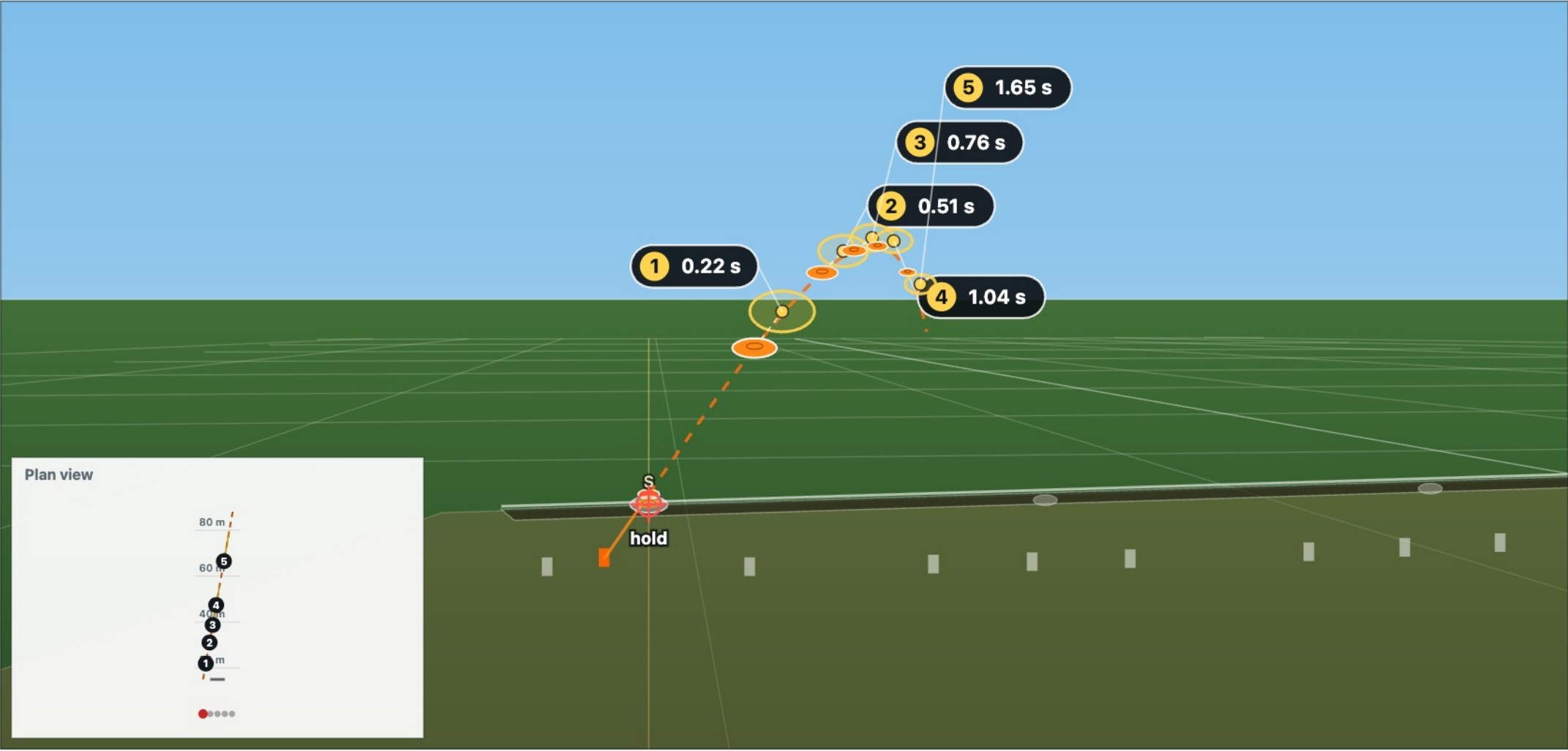
1 Release 0.22 s after release 87 cm ≈ 7.9 clays 74 cm right · 46 cm up +2.14° lateral · +1.32° vertical clay at 20 m · height 1.5 m · real separation 2.0 m first glimpse · within reach	2 Early 0.65 s after release 86 cm ≈ 7.8 clays 81 cm right · 28 cm up +1.57° lateral · +0.55° vertical clay at 30 m · height 3.5 m · real separation 3.2 m quick shot · within reach	3 Good window 1.01 s after release 89 cm ≈ 8.1 clays 89 cm right · 3 cm up +1.33° lateral · +0.04° vertical clay at 38 m · height 4.6 m · real separation 4.5 m the comfortable shot · at the limit	4 At the limit 1.41 s after release 106 cm ≈ 9.6 clays 98 cm right · 41 cm down +1.17° lateral · -0.50° vertical clay at 48 m · height 5.2 m · real separation 6.2 m last useful shot · at the limit	5 You are late 2.20 s after release 218 cm ≈ 19.8 clays 114 cm right · 186 cm down +0.99° lateral · -1.60° vertical clay at 66 m · height 4.5 m · real separation 10.2 m no longer reaching well · beyond useful range
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Station 1 · Machine 2

angle +10° · height 1.5 m at 10 m · fall 76 m · 116 km/h · elevation 8.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned +4° horizontally and +4° vertically



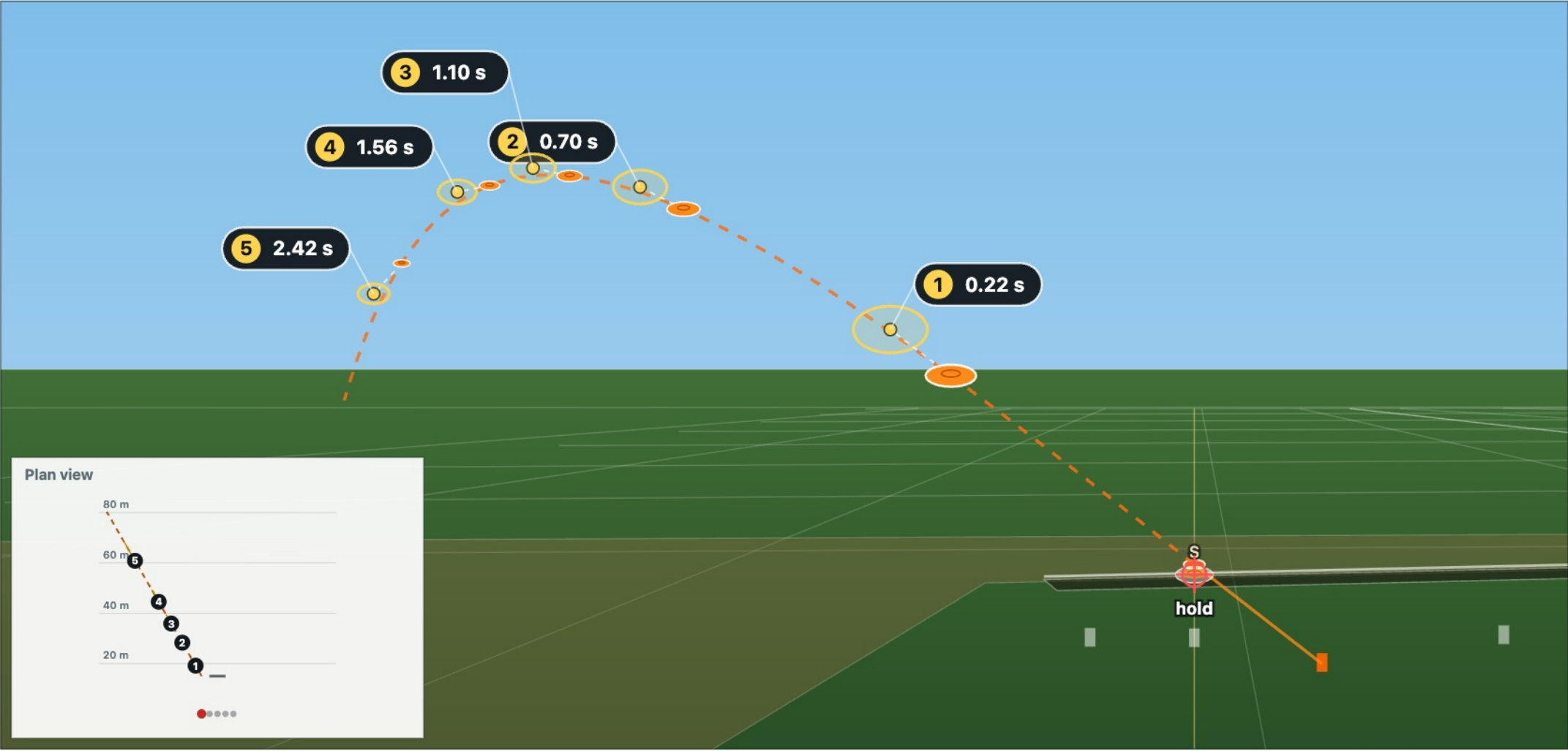
1 Release 0.22 s after release 57 cm ≈ 5.2 clays 36 cm right · 44 cm up +0.94° lateral · +1.15° vertical clay at 22 m · height 1.1 m · real separation 3.1 m first glimpse · within reach	2 Early 0.51 s after release 50 cm ≈ 4.6 clays 40 cm right · 31 cm up +0.73° lateral · +0.56° vertical clay at 31 m · height 2.1 m · real separation 4.8 m quick shot · within reach	3 Good window 0.76 s after release 45 cm ≈ 4.1 clays 43 cm right · 11 cm up +0.64° lateral · +0.17° vertical clay at 39 m · height 2.7 m · real separation 6.5 m the comfortable shot · at the limit	4 At the limit 1.04 s after release 53 cm ≈ 4.8 clays 48 cm right · 22 cm down +0.57° lateral · -0.27° vertical clay at 48 m · height 3.0 m · real separation 8.8 m last useful shot · at the limit	5 You are late 1.65 s after release 144 cm ≈ 13.1 clays 52 cm right · 134 cm down +0.44° lateral · -1.14° vertical clay at 67 m · height 2.7 m · real separation 13.4 m no longer reaching well · beyond useful range
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Station 1 · Machine 3

angle -30° · height 3.0 m at 10 m · fall 76 m · 83 km/h · elevation 18.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned -12° horizontally and +6° vertically



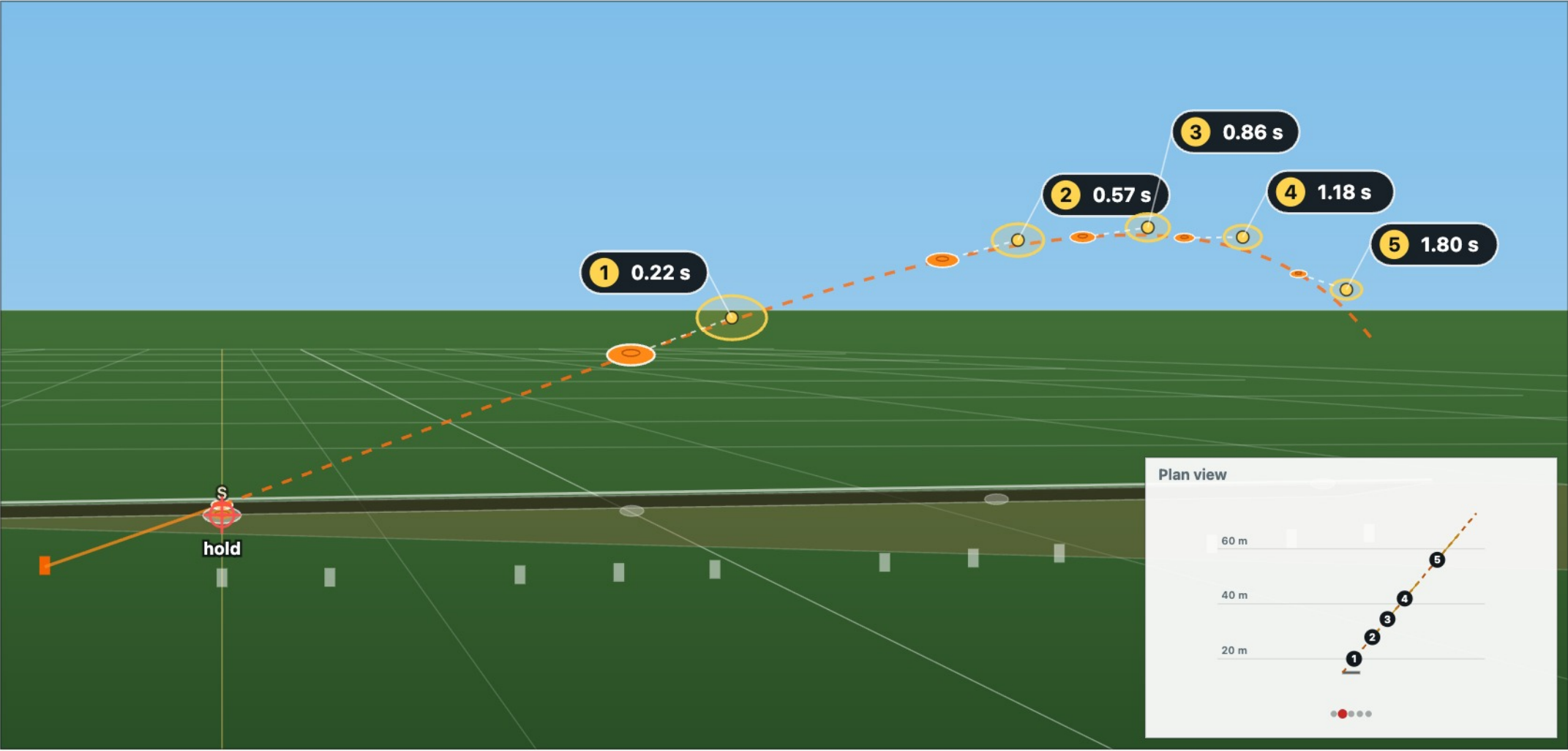
1 Release 0.22 s after release 82 cm ≈ 7.4 clays 66 cm left · 48 cm up -1.96° lateral · +1.42° vertical clay at 19 m · height 1.6 m · real separation 1.8 m first glimpse · within reach	2 Early 0.70 s after release 78 cm ≈ 7.1 clays 73 cm left · 29 cm up -1.42° lateral · +0.56° vertical clay at 29 m · height 4.2 m · real separation 2.9 m quick shot · within reach	3 Good window 1.10 s after release 79 cm ≈ 7.2 clays 79 cm left · 1 cm up -1.20° lateral · +0.01° vertical clay at 38 m · height 5.5 m · real separation 4.0 m the comfortable shot · at the limit	4 At the limit 1.56 s after release 100 cm ≈ 9.1 clays 88 cm left · 48 cm down -1.05° lateral · -0.57° vertical clay at 48 m · height 6.3 m · real separation 5.6 m last useful shot · at the limit	5 You are late 2.42 s after release 230 cm ≈ 20.9 clays 104 cm left · 206 cm down -0.89° lateral · -1.77° vertical clay at 66 m · height 5.4 m · real separation 9.3 m no longer reaching well · beyond useful range
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Station 2 · Machine 4

angle +40° · height 1.7 m at 10 m · fall 76 m · 108 km/h · elevation 10.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned +16° horizontally and +4° vertically



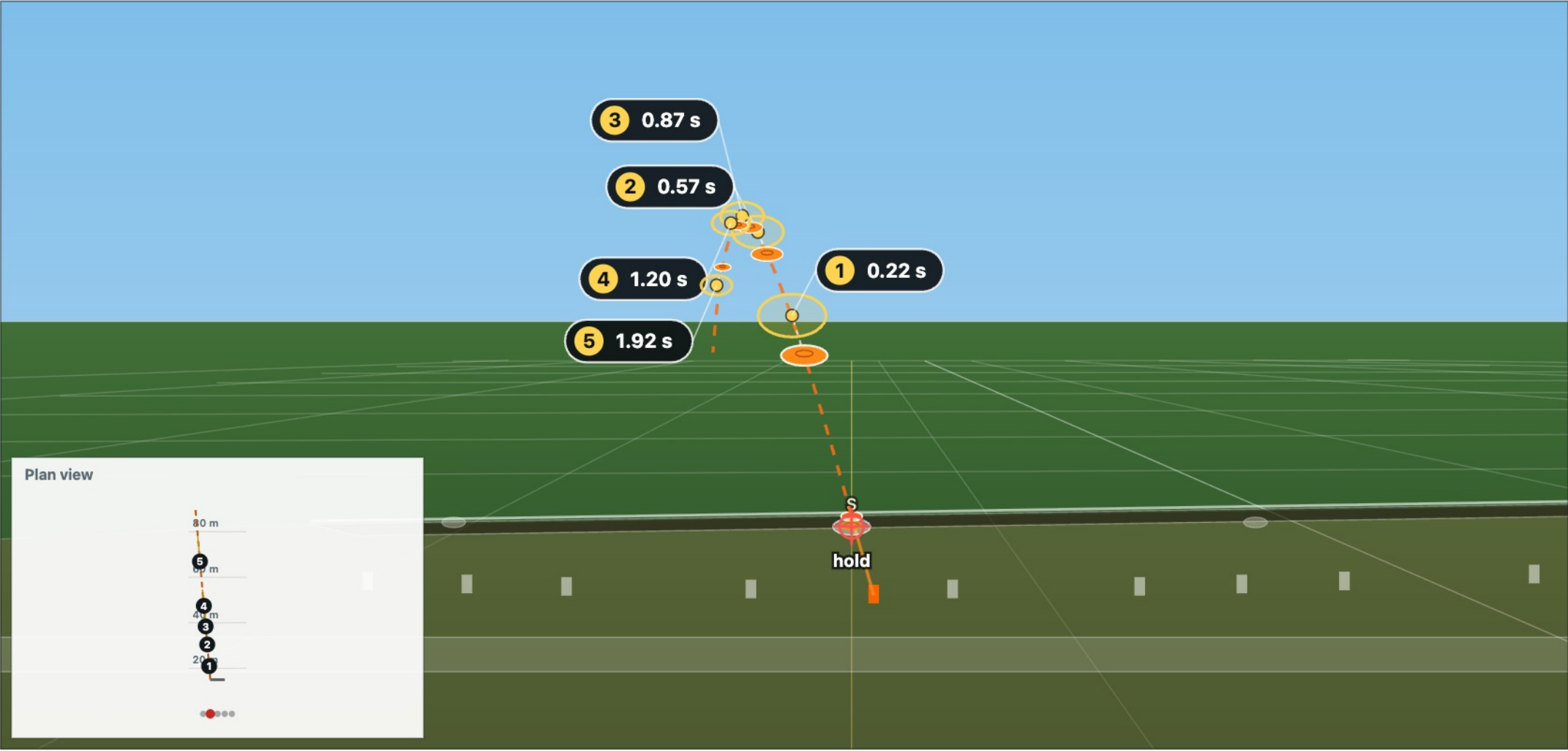
1 Release 0.22 s after release 127 cm ≈ 11.5 clays 120 cm right · 41 cm up +3.36° lateral · +1.15° vertical clay at 20 m · height 1.2 m · real separation 2.6 m first glimpse · within reach	2 Early 0.57 s after release 135 cm ≈ 12.3 clays 133 cm right · 26 cm up +2.54° lateral · +0.51° vertical clay at 30 m · height 2.5 m · real separation 4.1 m quick shot · within reach	3 Good window 0.86 s after release 145 cm ≈ 13.2 clays 145 cm right · 5 cm up +2.18° lateral · +0.07° vertical clay at 38 m · height 3.2 m · real separation 5.7 m the comfortable shot · at the limit	4 At the limit 1.18 s after release 165 cm ≈ 15.0 clays 162 cm right · 34 cm down +1.94° lateral · -0.41° vertical clay at 48 m · height 3.5 m · real separation 8.0 m last useful shot · at the limit	5 You are late 1.80 s after release 237 cm ≈ 21.5 clays 183 cm right · 151 cm down +1.59° lateral · -1.32° vertical clay at 66 m · height 2.9 m · real separation 12.5 m no longer reaching well · beyond useful range
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Station 2 · Machine 5

angle -5° · height 2.0 m at 10 m · fall 76 m · 100 km/h · elevation 12.0°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned -2° horizontally and +5° vertically



Plan view



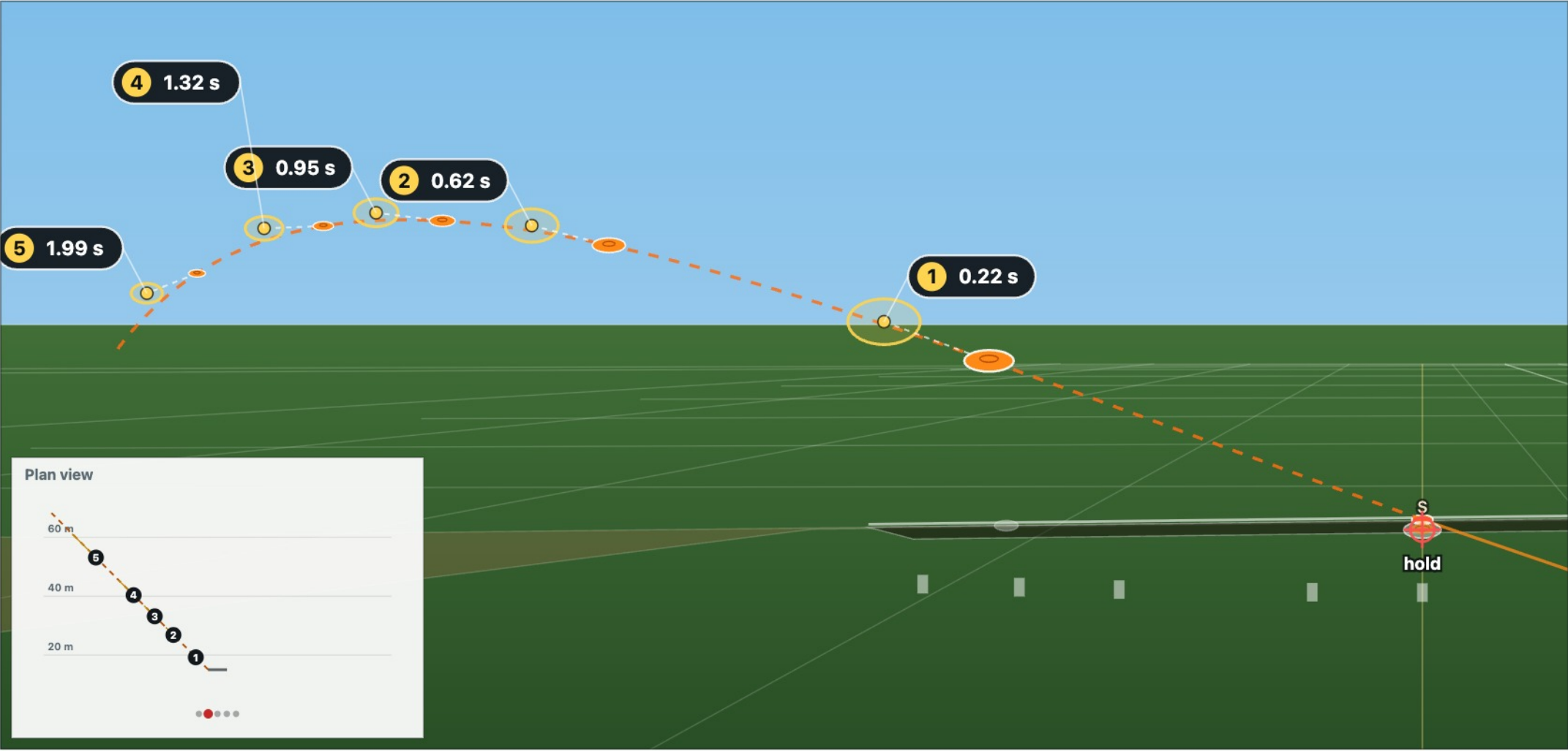
1 Release 0.22 s after release 48 cm ≈ 4.4 clays 15 cm left · 46 cm up -0.41° lateral · +1.25° vertical clay at 21 m · height 1.3 m · real separation 2.4 m first glimpse · within reach	2 Early 0.57 s after release 35 cm ≈ 3.1 clays 16 cm left · 30 cm up -0.31° lateral · +0.57° vertical clay at 31 m · height 2.8 m · real separation 3.9 m quick shot · within reach	3 Good window 0.87 s after release 20 cm ≈ 1.8 clays 18 cm left · 9 cm up -0.27° lateral · +0.13° vertical clay at 39 m · height 3.6 m · real separation 5.3 m the comfortable shot · at the limit	4 At the limit 1.20 s after release 35 cm ≈ 3.2 clays 20 cm left · 29 cm down -0.24° lateral · -0.35° vertical clay at 48 m · height 4.1 m · real separation 7.2 m last useful shot · at the limit	5 You are late 1.92 s after release 161 cm ≈ 14.6 clays 22 cm left · 159 cm down -0.19° lateral · -1.36° vertical clay at 67 m · height 3.6 m · real separation 11.5 m no longer reaching well · beyond useful range
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Station 2 · Machine 6

angle -45° · height 2.0 m at 10 m · fall 76 m · 100 km/h · elevation 12.0°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned -19° horizontally and +5° vertically



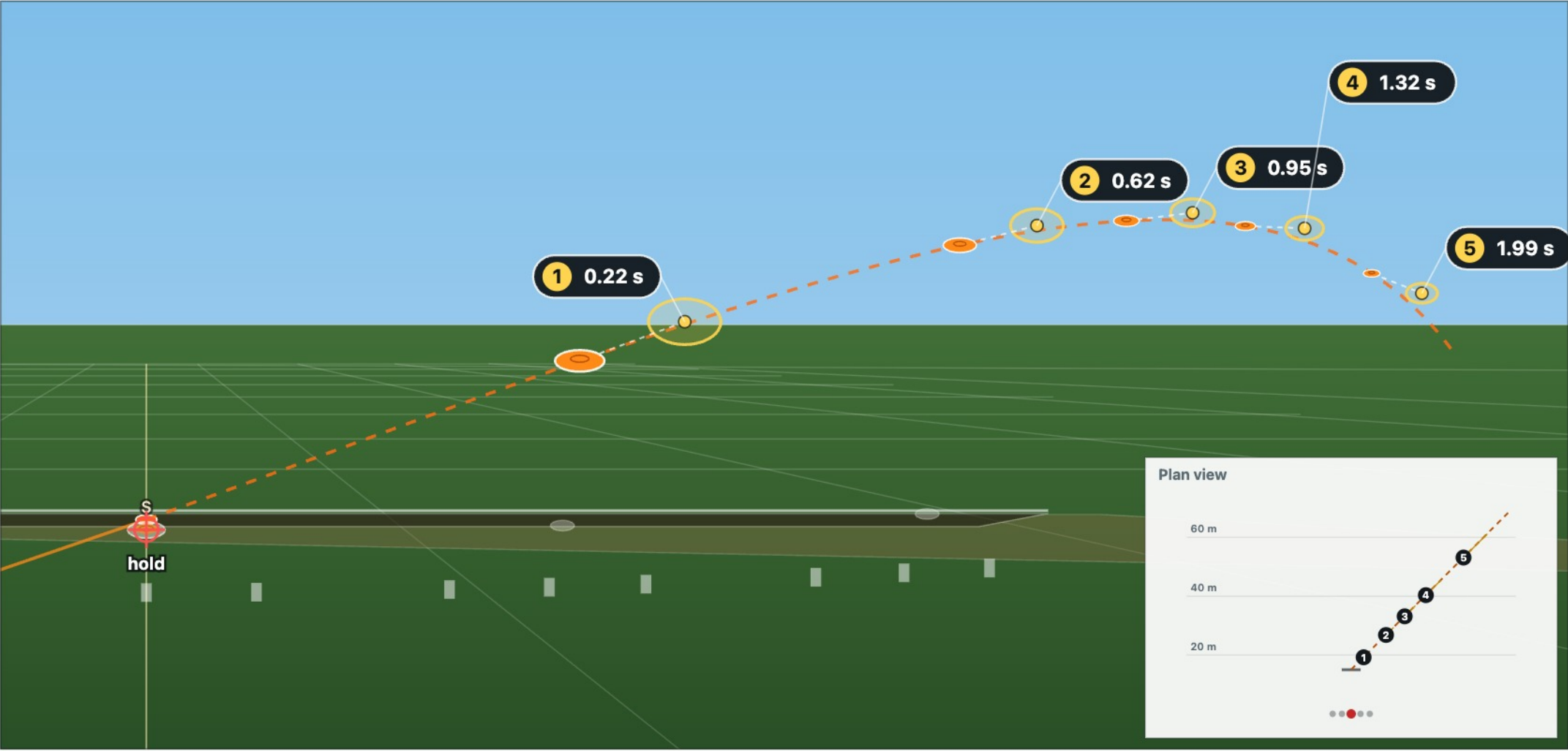
1 Release 0.22 s after release 125 cm ≈ 11.4 clays 118 cm left · 41 cm up -3.43° lateral · +1.20° vertical clay at 20 m · height 1.3 m · real separation 2.2 m first glimpse · within reach	2 Early 0.62 s after release 133 cm ≈ 12.1 clays 131 cm left · 25 cm up -2.55° lateral · +0.49° vertical clay at 29 m · height 2.9 m · real separation 3.6 m quick shot · within reach	3 Good window 0.95 s after release 143 cm ≈ 13.0 clays 143 cm left · 1 cm up -2.16° lateral · +0.01° vertical clay at 38 m · height 3.8 m · real separation 5.1 m the comfortable shot · within reach	4 At the limit 1.32 s after release 165 cm ≈ 15.0 clays 159 cm left · 42 cm down -1.91° lateral · -0.50° vertical clay at 48 m · height 4.2 m · real separation 7.2 m last useful shot · at the limit	5 You are late 1.99 s after release 250 cm ≈ 22.8 clays 185 cm left · 169 cm down -1.62° lateral · -1.48° vertical clay at 65 m · height 3.5 m · real separation 11.5 m no longer reaching well · beyond useful range
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Station 3 • Machine 7

angle +45° • height 2.0 m at 10 m • fall 76 m • 100 km/h • elevation 12.0°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 2.26x over the base field of 48° vertical • view turned +19° horizontally and +5° vertically



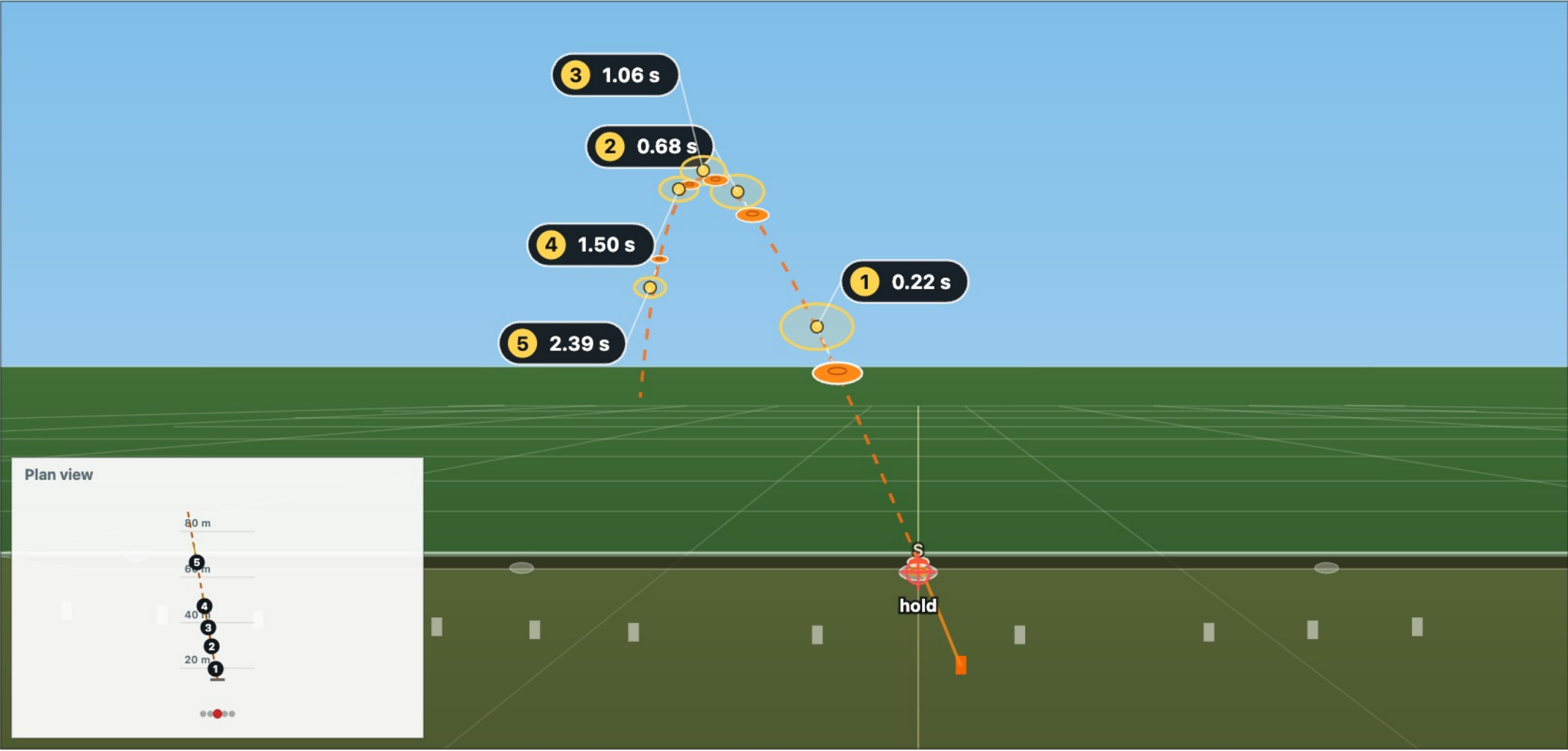
1 Release 0.22 s after release 125 cm ≈ 11.4 clays 118 cm right • 41 cm up +3.43° lateral • +1.20° vertical clay at 20 m • height 1.3 m • real separation 2.2 m first glimpse • within reach	2 Early 0.62 s after release 133 cm ≈ 12.1 clays 131 cm right • 25 cm up +2.55° lateral • +0.49° vertical clay at 29 m • height 2.9 m • real separation 3.6 m quick shot • within reach	3 Good window 0.95 s after release 143 cm ≈ 13.0 clays 143 cm right • 1 cm up +2.16° lateral • +0.01° vertical clay at 38 m • height 3.8 m • real separation 5.1 m the comfortable shot • within reach	4 At the limit 1.32 s after release 165 cm ≈ 15.0 clays 159 cm right • 42 cm down +1.91° lateral • -0.50° vertical clay at 48 m • height 4.2 m • real separation 7.2 m last useful shot • at the limit	5 You are late 1.99 s after release 250 cm ≈ 22.8 clays 185 cm right • 169 cm down +1.62° lateral • -1.48° vertical clay at 65 m • height 3.5 m • real separation 11.5 m no longer reaching well • beyond useful range
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Station 3 · Machine 8

angle -10° · height 3.0 m at 10 m · fall 76 m · 83 km/h · elevation 18.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned -4° horizontally and +6° vertically



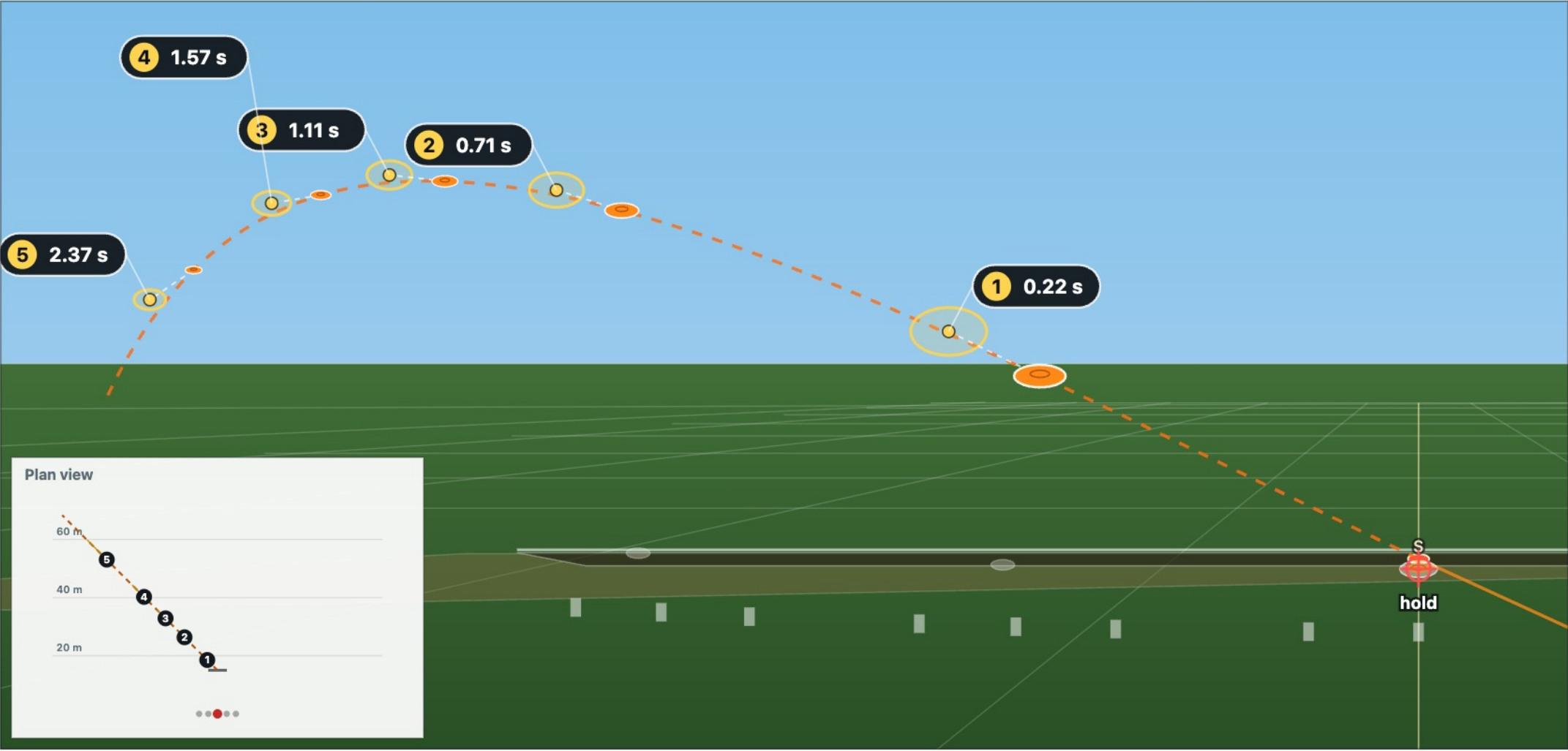
1 Release 0.22 s after release 55 cm ≈ 5.0 clays 23 cm left · 49 cm up -0.67° lateral · +1.43° vertical clay at 20 m · height 1.6 m · real separation 1.8 m first glimpse · within reach	2 Early 0.68 s after release 40 cm ≈ 3.7 clays 26 cm left · 31 cm up -0.49° lateral · +0.60° vertical clay at 30 m · height 4.1 m · real separation 3.0 m quick shot · within reach	3 Good window 1.06 s after release 28 cm ≈ 2.6 clays 28 cm left · 4 cm up -0.41° lateral · +0.07° vertical clay at 38 m · height 5.4 m · real separation 4.1 m the comfortable shot · at the limit	4 At the limit 1.50 s after release 52 cm ≈ 4.7 clays 30 cm left · 42 cm down -0.36° lateral · -0.50° vertical clay at 48 m · height 6.3 m · real separation 5.6 m last useful shot · at the limit	5 You are late 2.39 s after release 204 cm ≈ 18.6 clays 36 cm left · 201 cm down -0.30° lateral · -1.71° vertical clay at 67 m · height 5.5 m · real separation 9.3 m no longer reaching well · beyond useful range
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Station 3 • Machine 9

angle -45° • height 2.8 m at 10 m • fall 76 m • 85 km/h • elevation 16.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 2.26x over the base field of 48° vertical • view turned -18° horizontally and +6° vertically



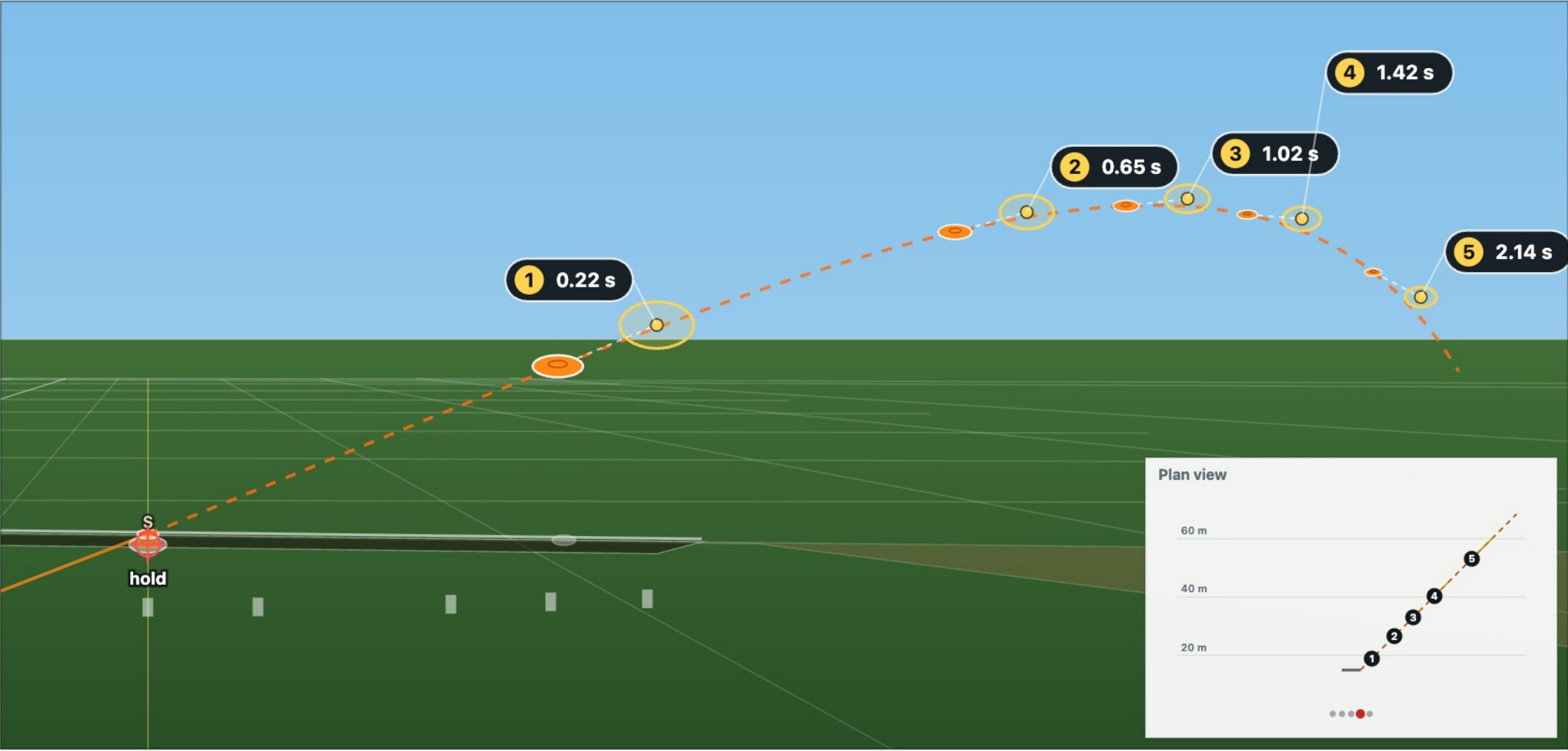
1 Release 0.22 s after release 106 cm ≈ 9.6 clays 96 cm left • 45 cm up -2.91° lateral • +1.36° vertical clay at 19 m • height 1.6 m • real separation 1.8 m first glimpse • within reach	2 Early 0.71 s after release 109 cm ≈ 9.9 clays 106 cm left • 26 cm up -2.12° lateral • +0.51° vertical clay at 29 m • height 4.0 m • real separation 2.9 m quick shot • within reach	3 Good window 1.11 s after release 116 cm ≈ 10.6 clays 116 cm left • 3 cm down -1.77° lateral • -0.04° vertical clay at 38 m • height 5.3 m • real separation 4.1 m the comfortable shot • within reach	4 At the limit 1.57 s after release 139 cm ≈ 12.6 clays 129 cm left • 53 cm down -1.54° lateral • -0.63° vertical clay at 48 m • height 5.9 m • real separation 5.8 m last useful shot • at the limit	5 You are late 2.37 s after release 256 cm ≈ 23.3 clays 155 cm left • 204 cm down -1.35° lateral • -1.78° vertical clay at 65 m • height 4.9 m • real separation 9.7 m no longer reaching well • beyond useful range
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Station 4 · Machine 10

angle +45° · height 2.3 m at 10 m · fall 76 m · 93 km/h · elevation 13.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned +18° horizontally and +5° vertically



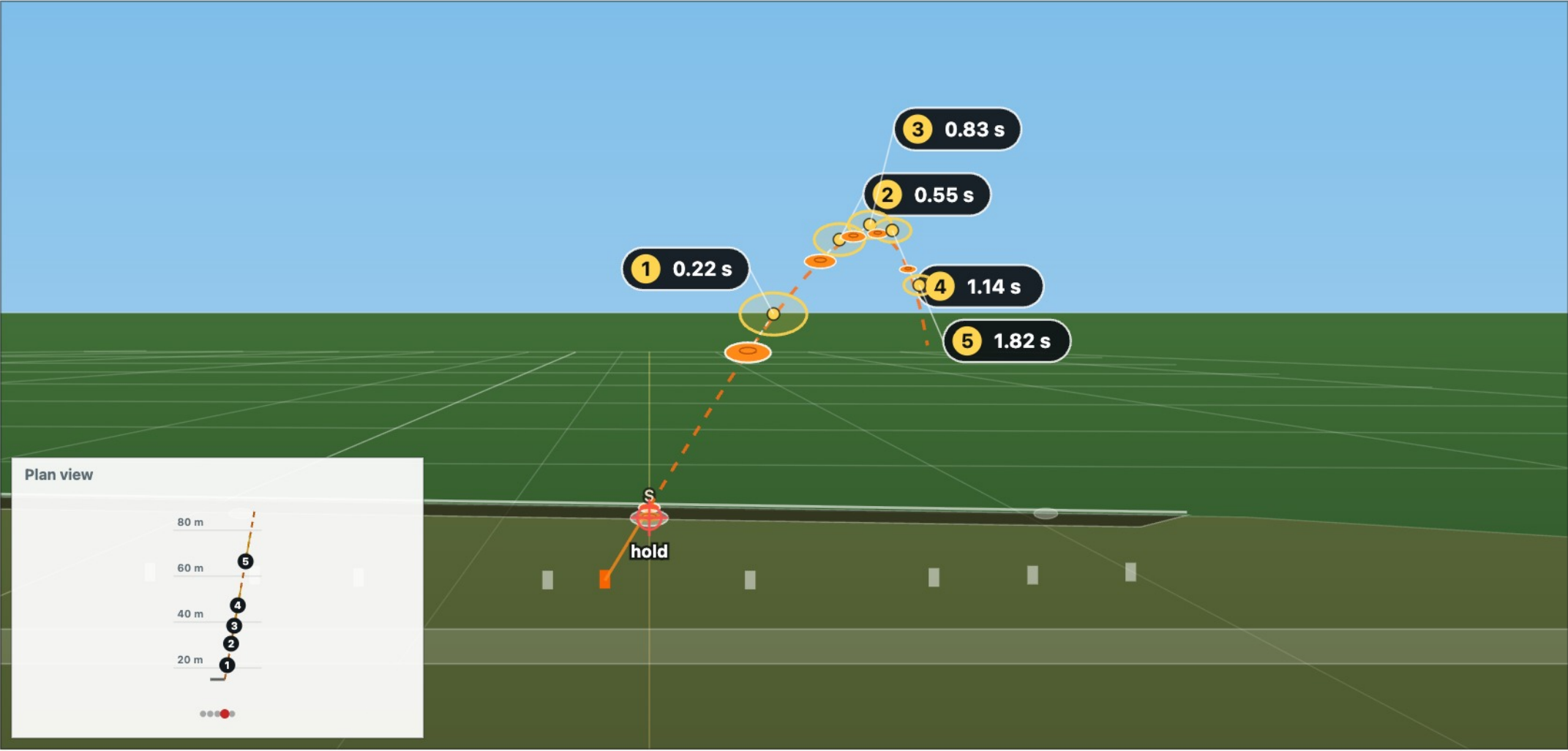
1 Release 0.22 s after release 116 cm ≈ 10.6 clays 108 cm right · 43 cm up +3.20° lateral · +1.26° vertical clay at 19 m · height 1.4 m · real separation 2.0 m first glimpse · within reach	2 Early 0.65 s after release 122 cm ≈ 11.1 clays 120 cm right · 25 cm up +2.36° lateral · +0.49° vertical clay at 29 m · height 3.3 m · real separation 3.3 m quick shot · within reach	3 Good window 1.02 s after release 131 cm ≈ 11.9 clays 131 cm right · 1 cm down +1.99° lateral · -0.01° vertical clay at 38 m · height 4.3 m · real separation 4.7 m the comfortable shot · within reach	4 At the limit 1.42 s after release 153 cm ≈ 13.9 clays 146 cm right · 47 cm down +1.74° lateral · -0.56° vertical clay at 48 m · height 4.8 m · real separation 6.5 m last useful shot · at the limit	5 You are late 2.14 s after release 251 cm ≈ 22.8 clays 171 cm right · 183 cm down +1.50° lateral · -1.60° vertical clay at 65 m · height 4.0 m · real separation 10.7 m no longer reaching well · beyond useful range
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Station 4 · Machine 11

angle +10° · height 1.8 m at 10 m · fall 76 m · 105 km/h · elevation 10.7°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned +4° horizontally and +4° vertically



Plan view



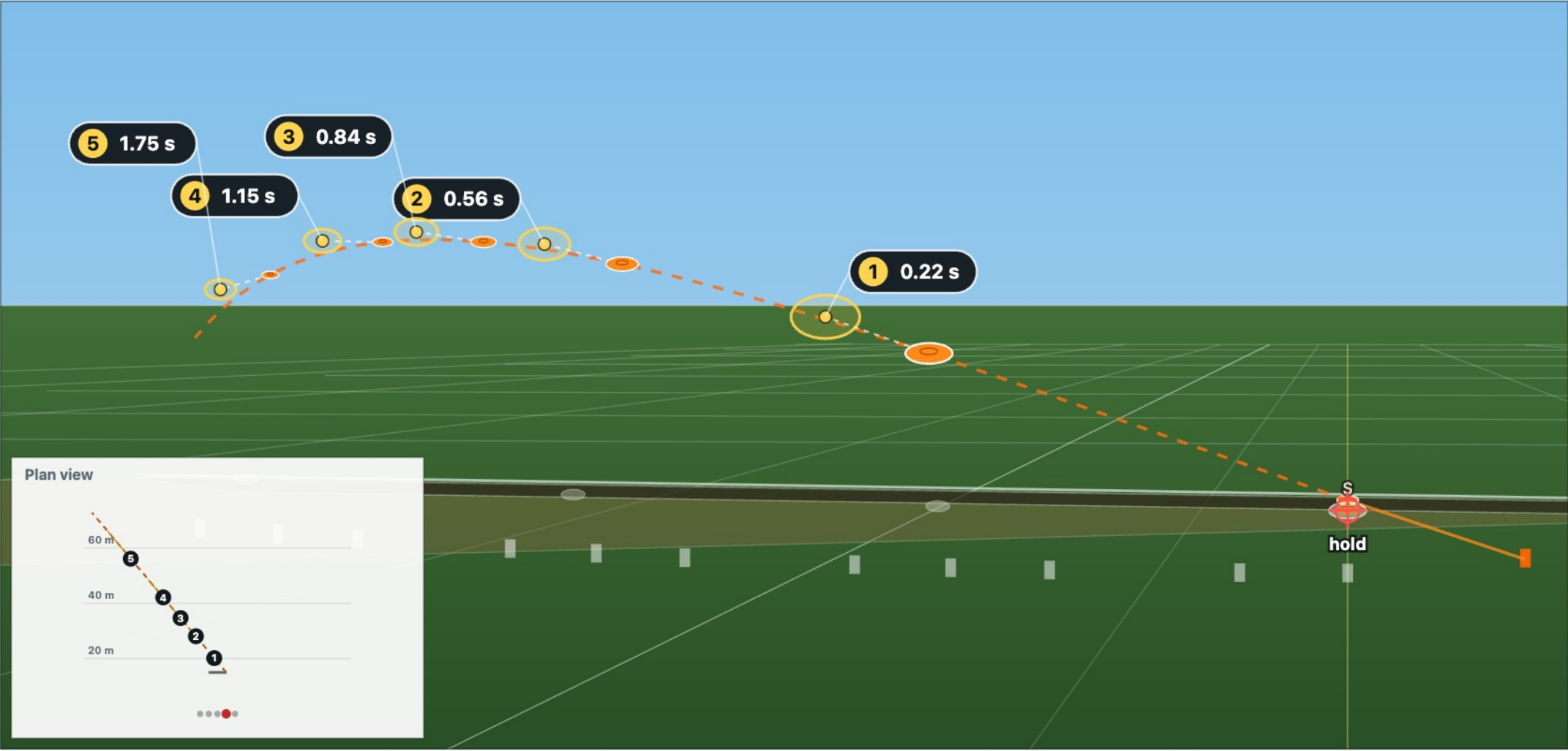
1 Release 0.22 s after release 55 cm ≈ 5.0 clays 32 cm right · 45 cm up +0.86° lateral · +1.21° vertical clay at 21 m · height 1.3 m · real separation 2.6 m first glimpse · within reach	2 Early 0.55 s after release 46 cm ≈ 4.2 clays 35 cm right · 30 cm up +0.65° lateral · +0.56° vertical clay at 31 m · height 2.5 m · real separation 4.2 m quick shot · within reach	3 Good window 0.83 s after release 39 cm ≈ 3.6 clays 38 cm right · 9 cm up +0.57° lateral · +0.14° vertical clay at 39 m · height 3.2 m · real separation 5.7 m the comfortable shot · at the limit	4 At the limit 1.14 s after release 50 cm ≈ 4.6 clays 42 cm right · 27 cm down +0.51° lateral · -0.32° vertical clay at 48 m · height 3.7 m · real separation 7.7 m last useful shot · at the limit	5 You are late 1.82 s after release 157 cm ≈ 14.3 clays 47 cm right · 150 cm down +0.40° lateral · -1.28° vertical clay at 67 m · height 3.2 m · real separation 12.1 m no longer reaching well · beyond useful range
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Station 4 · Machine 12

angle -40° · height 1.6 m at 10 m · fall 76 m · 112 km/h · elevation 9.4°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned -16° horizontally and +4° vertically



Plan view



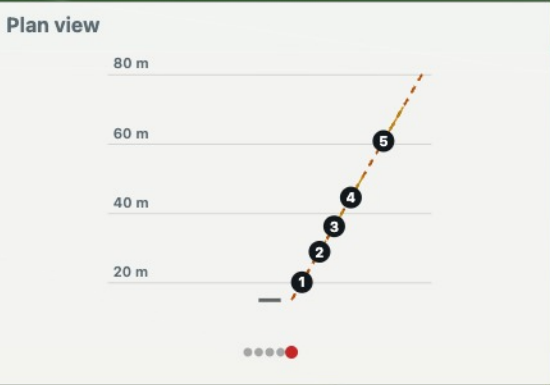
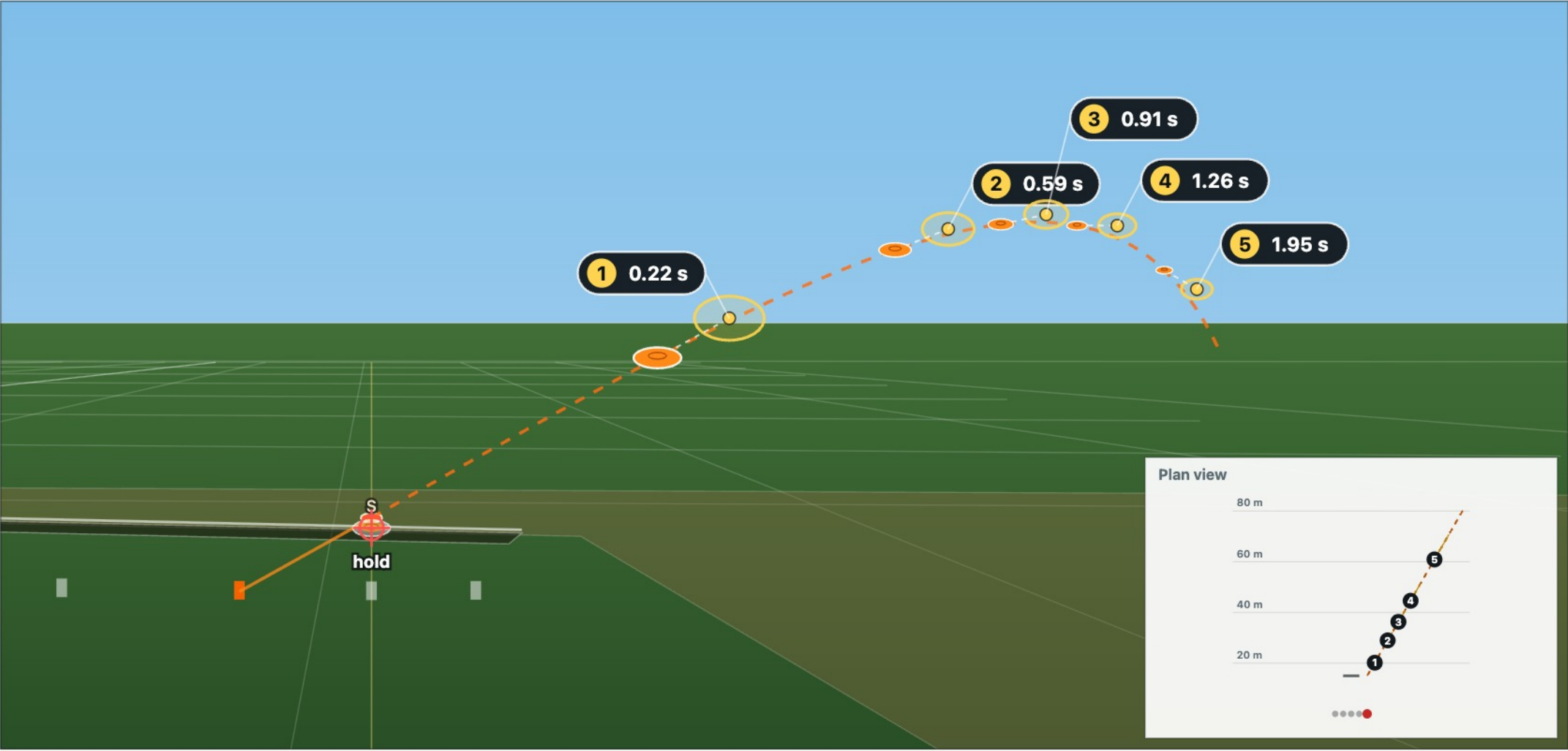
1 Release 0.22 s after release 131 cm ≈ 11.9 clays 125 cm left · 41 cm up -3.46° lateral · +1.13° vertical clay at 21 m · height 1.2 m · real separation 2.7 m first glimpse · within reach	2 Early 0.56 s after release 141 cm ≈ 12.8 clays 138 cm left · 26 cm up -2.63° lateral · +0.50° vertical clay at 30 m · height 2.4 m · real separation 4.3 m quick shot · within reach	3 Good window 0.84 s after release 152 cm ≈ 13.8 clays 151 cm left · 5 cm up -2.26° lateral · +0.07° vertical clay at 38 m · height 3.0 m · real separation 6.0 m the comfortable shot · at the limit	4 At the limit 1.15 s after release 172 cm ≈ 15.7 clays 169 cm left · 34 cm down -2.02° lateral · -0.40° vertical clay at 48 m · height 3.3 m · real separation 8.4 m last useful shot · at the limit	5 You are late 1.75 s after release 238 cm ≈ 21.7 clays 189 cm left · 146 cm down -1.64° lateral · -1.27° vertical clay at 66 m · height 2.8 m · real separation 12.9 m no longer reaching well · beyond useful range
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Station 5 · Machine 13

angle +30° · height 2.0 m at 10 m · fall 76 m · 100 km/h · elevation 12.0°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned +12° horizontally and +5° vertically



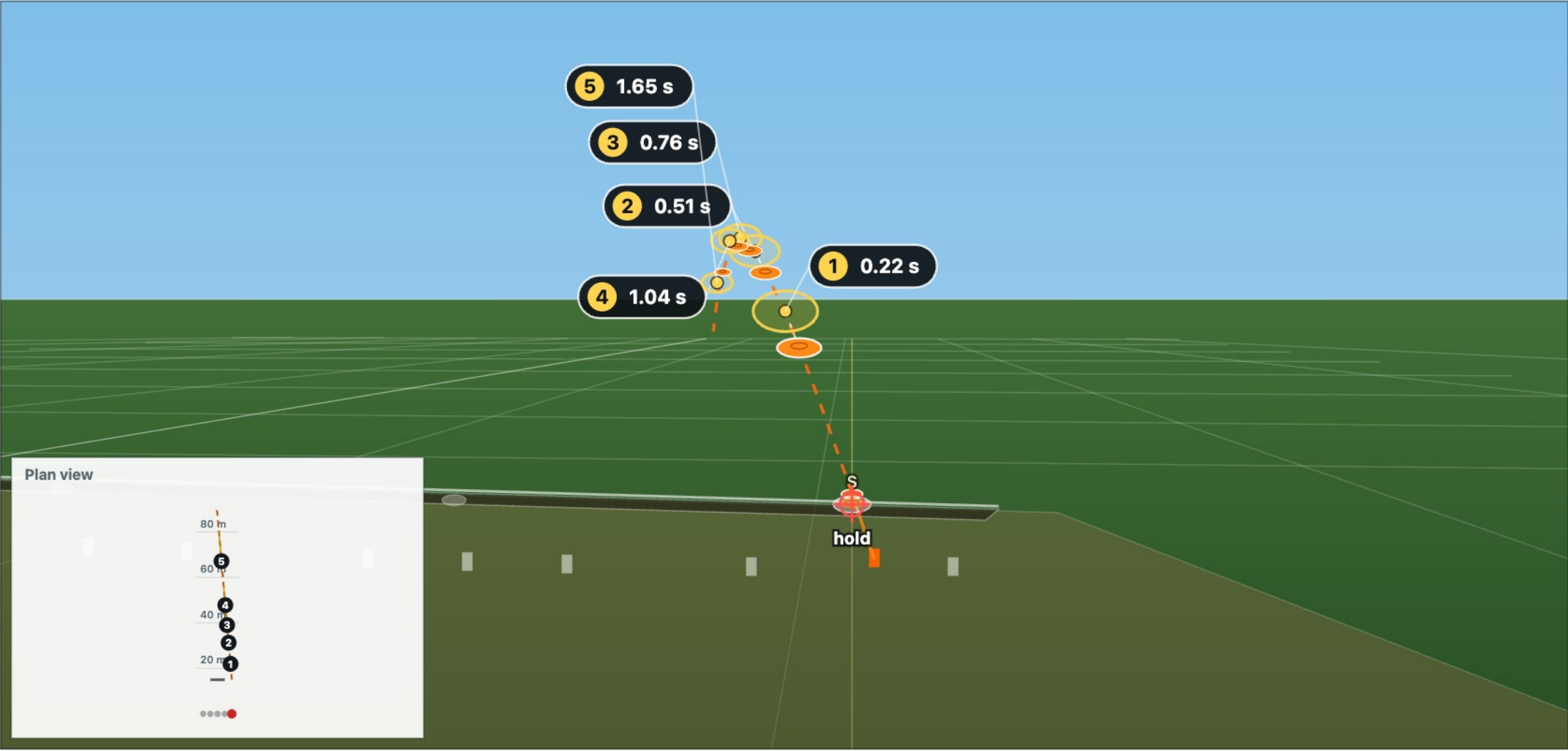
1 Release 0.22 s after release 95 cm ≈ 8.7 clays 85 cm right · 44 cm up +2.38° lateral · +1.23° vertical clay at 20 m · height 1.3 m · real separation 2.3 m first glimpse · within reach	2 Early 0.59 s after release 98 cm ≈ 8.9 clays 94 cm right · 28 cm up +1.78° lateral · +0.53° vertical clay at 30 m · height 2.9 m · real separation 3.8 m quick shot · within reach	3 Good window 0.91 s after release 102 cm ≈ 9.3 clays 102 cm right · 5 cm up +1.52° lateral · +0.07° vertical clay at 38 m · height 3.7 m · real separation 5.2 m the comfortable shot · at the limit	4 At the limit 1.26 s after release 119 cm ≈ 10.8 clays 113 cm right · 36 cm down +1.35° lateral · -0.43° vertical clay at 48 m · height 4.1 m · real separation 7.2 m last useful shot · at the limit	5 You are late 1.95 s after release 208 cm ≈ 18.9 clays 129 cm right · 164 cm down +1.11° lateral · -1.41° vertical clay at 66 m · height 3.5 m · real separation 11.5 m no longer reaching well · beyond useful range
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Station 5 · Machine 14

angle -5° · height 1.5 m at 10 m · fall 76 m · 116 km/h · elevation 8.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.26x over the base field of 48° vertical · view turned -2° horizontally and +4° vertically



Plan view



1 Release
0.22 s after release

48 cm

≈ 4.4 clays

18 cm left · 44 cm up

-0.47° lateral · +1.15° vertical

clay at 22 m · height 1.1 m · real separation 3.1 m

first glimpse · within reach

2 Early
0.51 s after release

37 cm

≈ 3.3 clays

20 cm left · 31 cm up

-0.37° lateral · +0.56° vertical

clay at 31 m · height 2.1 m · real separation 4.8 m

quick shot · within reach

3 Good window
0.76 s after release

25 cm

≈ 2.2 clays

22 cm left · 11 cm up

-0.32° lateral · +0.17° vertical

clay at 39 m · height 2.7 m · real separation 6.5 m

the comfortable shot · at the limit

4 At the limit
1.04 s after release

33 cm

≈ 3.0 clays

24 cm left · 23 cm down

-0.29° lateral · -0.27° vertical

clay at 48 m · height 3.0 m · real separation 8.8 m

last useful shot · at the limit

5 You are late
1.65 s after release

136 cm

≈ 12.4 clays

26 cm left · 134 cm down

-0.22° lateral · -1.14° vertical

clay at 67 m · height 2.7 m · real separation 13.4 m

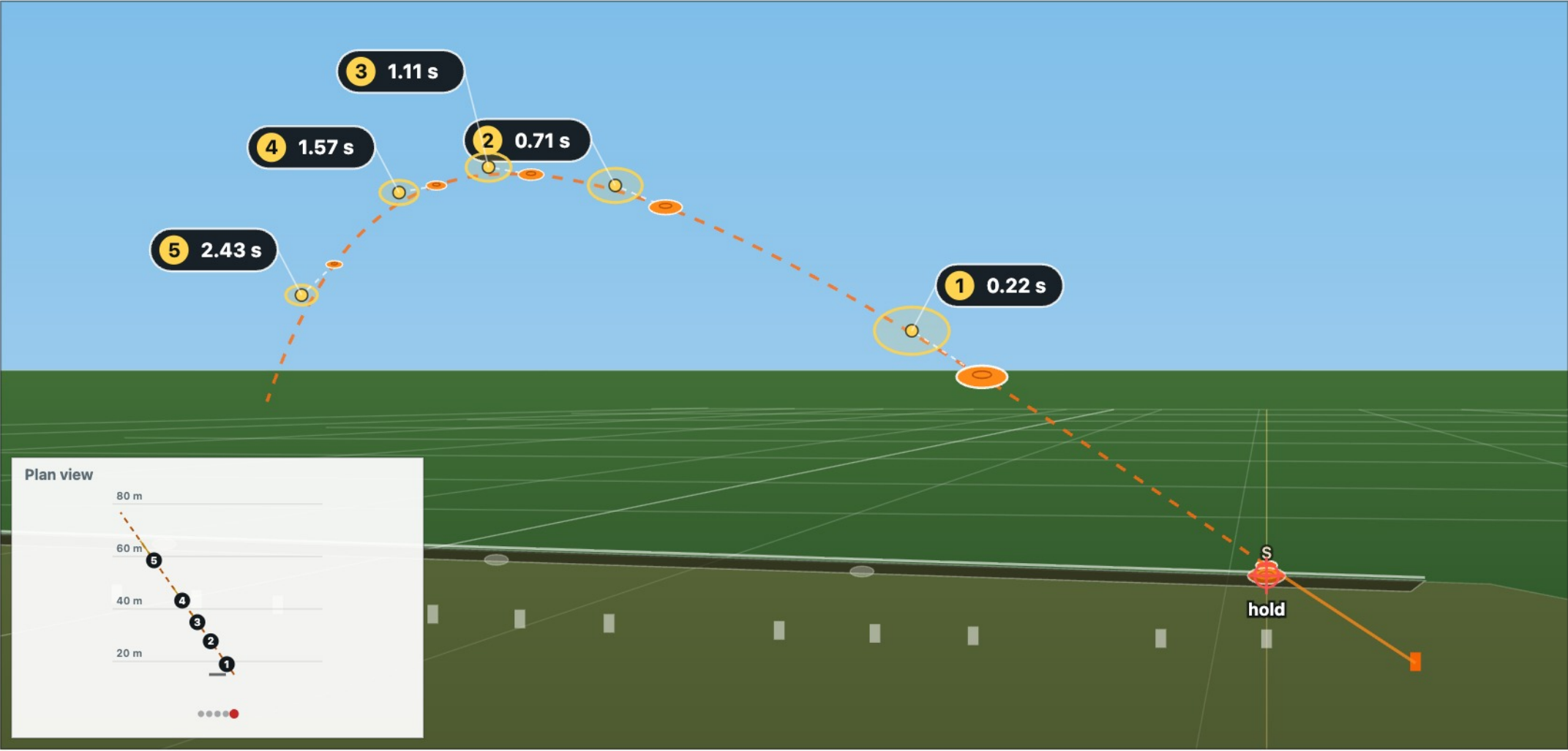
no longer reaching well · beyond useful range

Station 5 • Machine 15

angle -35° • height 3.0 m at 10 m • fall 76 m • 83 km/h • elevation 18.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 2.26x over the base field of 48° vertical • view turned -14° horizontally and +6° vertically



1 Release 0.22 s after release 89 cm ≈ 8.1 clays 75 cm left • 47 cm up -2.26° lateral • +1.41° vertical clay at 19 m • height 1.6 m • real separation 1.7 m first glimpse • within reach	2 Early 0.71 s after release 88 cm ≈ 8.0 clays 83 cm left • 28 cm up -1.64° lateral • +0.55° vertical clay at 29 m • height 4.2 m • real separation 2.9 m quick shot • within reach	3 Good window 1.11 s after release 91 cm ≈ 8.3 clays 91 cm left • 0 cm down -1.38° lateral • -0.00° vertical clay at 38 m • height 5.6 m • real separation 4.0 m the comfortable shot • within reach	4 At the limit 1.57 s after release 112 cm ≈ 10.1 clays 100 cm left • 49 cm down -1.20° lateral • -0.59° vertical clay at 48 m • height 6.3 m • real separation 5.5 m last useful shot • at the limit	5 You are late 2.43 s after release 239 cm ≈ 21.7 clays 120 cm left • 207 cm down -1.04° lateral • -1.79° vertical clay at 66 m • height 5.4 m • real separation 9.3 m no longer reaching well • beyond useful range
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Notice of use and licence

Read it before applying anything from this notebook on a shooting ground

This is an estimate, not a measurement

The trajectories, the times and the leads come from a simplified model that solves a visual parabola from the height and the range of the clay, with reduced gravity and a common visual release. It does not measure a real trap, it does not reproduce the full aerodynamics of the disc, and it does not replace a ballistic table for the cartridge you use.

The scheme tables reproduce values published by the federations, but this document does not certify any installation or any trajectory. In competition the authorised officials always decide.

The responsibility for the shot is yours

Use these figures as a starting point for observing and comparing, never as an instruction to follow blindly. Check your own numbers on the ground, with your gun, your cartridge and the conditions of the day, and correct whatever needs correcting.

This material does not replace in-person training, the rules of the ground, the help of an instructor or a gunsmith, or the legal requirements in force. Safety and shooting decisions are the responsibility of whoever pulls the trigger.

Before taking it to the ground

- Check that the scheme on the cover really is the one that is going to be thrown. Ask whoever is in charge instead of assuming it.
- The leads depend on the cartridge and the rib you set when generating the notebook. With different equipment, the figures change.
- Treat every number as a hypothesis you are going to test by shooting, not as the result you ought to get.

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Where everything is

Full book, simulator and source code:

www.robertocanales.com/proyectos#plato

rcanalescoder.github.io/Plato

rcanalesmora@gmail.com

The project is in beta

The book and the simulator are still under review and are shared so that anyone can try them, find errors and help improve them. If you find something that does not match your experience on the ground, the rules that apply to you, or the figures you measure yourself, write in: that kind of report is what makes the model better.